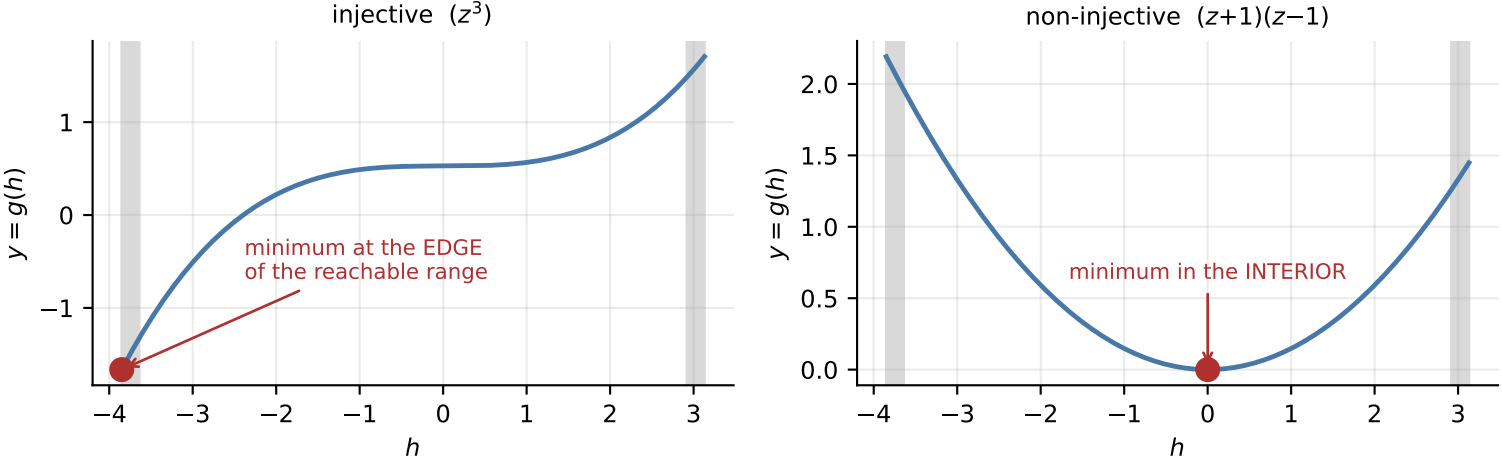A monotone g can only be minimised at an endpoint. So injectivity decides what the optimisation problem is.